

15. Uncertainty budget, limitations, and open questions

The smallest Z steps and the absolute bias are the measurements most limited by the setup. The relative measurements are robust: noise, rise distance, transfer-curve shape, and detection minimums. The magnitudes below are typical values, not measured ones; replace them with measured values after Step 4.

Source	Estimated magnitude	Affects	Mitigation
Robot repeatability	0.02–0.05 mm	Smallest B-Z steps; edge position	Approach every Z visit from the same direction; A, B, A, B averaging over the cycles; the ladder floored at the robot's resolvable move of 0.1 mm, rungs below twice the repeatability flagged as unreliable truth, and the ratio of the repeatability to each rung reported with δ_{50} ; the ramp probe needs no small Z moves at all; averaging over phase poses
Robot absolute accuracy	0.2–1 mm over large moves	bias(Z), registration	Plane-correspondence registration over 30 poses (REGISTRATION_POSES) spanning the full Z range; tests run as local moves from registered stations
Registration residual	≤ 0.15 mm RMS (acceptance limit)	bias(Z), edge offset s_{50} , small-feature area	Acceptance gate (Step 4.7); report the residual with every result
Plane-registration depth offset	about 0.08 mm at the indicative noise with 30 poses to $\pm 20^\circ$: the registration reports the standard error of the camera depth offset from its fit covariance, which replaces the earlier rough estimate (the plane-fit error is amplified by $1/(1 - \cos 20^\circ) \approx 17$ at the tilt limit)	Absolute bias(Z) only	Many poses over the full Z range; tilts to the $\pm 20^\circ$ limit (REGISTRATION_TILT_RANGE_DEG); the standard error reported with every result
As-built diameter uncertainty	set by the measuring instrument	A_{true} : relative error $\approx 2\delta D/D$, for example 0.6% at 7 mm with $\delta D = 0.02$ mm	Optical comparator or microscope; carry δD into the area and D_0 confidence intervals
Plate flatness	≤ 0.05 mm		Flatness spec; map the plate on a CMM if available

		Fixed-pattern noise, bias	
Thermal drift	unknown until the sentinels and the optional drift run have run	bias(Z), B-Z	Warm-up gate; sentinels on the mounted target, with the drift rate and the sensor-or-robot attribution per mount (Section 10, Step 11); the optional drift run of Section 4, Step 3; A, B, A, B ordering; randomized station order
Front-back interreflection	unknown	Cutout and small-gap results	Compare G small vs large; matte finish
Sensor temporal filter	depends on configuration	σ_t (underestimated), trial independence	Discard frames after each move until the plane mean settles (set in Step 4.4); run filters off if possible
Reflectance mismatch	avoided by design	Edges, area	One finish on all surfaces
Field coverage	achieved fraction below the requested 0.6 of the half field at the near stations (36% at 476 mm with the indicative geometry)	Field comparisons of A and C	The plan tool pulls off-axis targets inward to fit the field and logs the achieved fraction, which the A and C field comparisons carry as a factor
Tilt sub-series range	tilts only at 800 and 1600 mm with the 400 mm plate	Incidence-angle results (Section 10, Step 10)	Near-edge rule of Section 5, Step 5; a smaller plate or an added 566 mm tilt station extends the range
Quantizer dithering	expected at the near stations (noise about 0.6 quantum)	δZ_q and q from the ramp	The ramp reports whether the row average is stepped or smooth and takes the quantum from the depth levels when smooth (Section 11, Step 15); the optional staircase is a second pass
2-D image	not characterized	Any use of the 2-D image	Its visible area A_{geo_2D} is computed (Section 12, Step 7); brightness follows from the light and surface geometry, which the procedure records

Limitations. Results hold for one surface finish, static targets, mostly fronto-parallel poses, and the sensor configuration recorded in `sensor_config.json`. Registration from planes leaves the camera's depth offset conditioned by the registration tilt range (table above), so absolute bias(Z) carries that uncertainty; Z-scale and nonlinear bias do not, because a rigid transform cannot absorb them. The 0% point is an extrapolation of the fitted detection curve below the lowest measured point, `D_5`, and is reported as a prediction, not a measurement. Because the read-back robot pose is the step truth of B-Z, the smallest detectable Z step for the larger patches may come out as a bound set by the robot's repeatability, through the 0.1 mm rung floor and the truth rule of twice the repeatability, rather than a measurement. Where the frames dither the quantizer, the ramp's quantum comes from the spacing of the depth levels rather than from plateaus (Section 11, Step 15). Off-axis results at the near stations lie

closer to the axis than the requested field position, by the achieved fraction the plan tool logs. The tilt sub-series covers only 800 and 1600 mm with the 400 mm plate. The attribution of a drift to the sensor or to the robot and mounts needs the optional drift run of Section 4, Step 3. The 2-D image is not characterized beyond its computed visible area. The range limits 400 and 1600 mm assume the sensor reads there; Step 4.5 confirms this before any series runs.

Open questions

- ☐ VSX3000 f_x , f_y , baseline, projector offset, 2-D sensor position, FOV, depth LSB, and frame rate in the chosen trigger mode. These fill the $\dagger$ parameters, the chamfer check, and the time budget.
- ☐ Which on-sensor processing configuration is the production one?
- ☐ Which surface finish represents the production targets, and is a second finish needed?
- ☐ Does the sensor dither its disparity quantizer? The ramp of Section 6.2 answers it (Section 11, Step 15).
- ☐ Is 0.1 mm the smallest Z move the robot resolves (ROBOT_MIN_RESOLVABLE_MOVE_MM)? The optional staircase of Section 6.2 checks it.
- ☐ Is the sensor's thermal drift a reproducible function of its temperature reading? The optional drift run of Section 4, Step 3, compared with the in-session sentinels, answers it.